

Decent Coaching Classes
IX CBSE
Academic Conquest - III

Subject: Science

Marks: 30

Chapter 8: Journey Inside the Atom

Time: 1 Hour

General Instructions:

1. This question paper consists of four sections – A, B, C and D.
2. Section A has 6 objective-type questions of 1 mark each.
3. Section B has 4 very short answer questions of 2 marks each.
4. Section C has 4 short answer questions of 3 marks each. An internal choice is provided in one question.
5. Section D has 1 case-based/analytical question of 4 marks.
6. All questions are compulsory unless an internal choice is indicated.

SECTION - A

Multiple Choice Questions (6 x 1 = 6 Marks)

1. Who among the following discovered the electron through the study of cathode rays?
(a) John Dalton (b) J.J. Thomson (c) Ernest Rutherford (d) James Chadwick
2. Rutherford's model of the atom is also known as the:
(a) Plum pudding model (b) Planetary model (c) Quantum mechanical model (d) Stationary state model
3. The subatomic particle discovered by James Chadwick is the:
(a) Electron (b) Proton (c) Neutron (d) Positron
4. The maximum number of electrons that can be accommodated in the M-shell ($n = 3$) is:
(a) 2 (b) 8 (c) 18 (d) 32
5. Atoms of the same element having the same atomic number but different mass numbers are called:
(a) Isobars (b) Isotopes (c) Isomers (d) Ions
6. Assertion (A): In Rutherford's gold foil experiment, most of the alpha particles passed straight through the gold foil.
Reason (R): Most of the space inside an atom is empty.
Choose the correct option:
(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION - B

Very Short Answer Questions (4 x 2 = 8 Marks)

7. Define atomic number and mass number of an element.
8. State any two postulates of Bohr's model of the atom.
9. Write the electronic configuration of chlorine (atomic number 17). How many valence electrons does it have?
10. What is meant by the valency of an element? State the valency of sodium, whose electronic configuration is 2, 8, 1.

SECTION - C

Short Answer Questions (4 x 3 = 12 Marks)

11. Describe Rutherford's gold foil experiment. State any two observations he made.
12. Explain why Rutherford's atomic model could not explain the stability of an atom.

OR

Explain how Bohr's atomic model successfully accounted for the stability of the atom.

13. The electronic configuration of oxygen is 2, 6 and that of carbon is 2, 4. Using this information, deduce the valency of oxygen and of carbon, giving proper reasoning.
14. Bromine occurs in nature as two isotopes, ^{79}Br (49.7%) and ^{81}Br (50.3%). Calculate the average atomic mass of bromine.

SECTION - D

Case-based / Analytical Question (4 Marks)

15. Read the following data carefully and answer the questions that follow:

The composition of the nuclei of three atomic species X, Y and Z is given below:

	X	Y	Z
Number of protons	18	17	17
Number of neutrons	19	18	20

- (i) Calculate the mass number of X, Y and Z. (1 mark)
- (ii) What is the relation between Y and Z? Justify your answer. (1.5 marks)
- (iii) What is the relation between Z and X? Name such a pair of atoms. (1.5 marks)